

Spectral Reformulations and Partial Structure Results for a Problem of Green on Gowers Box Norms over Finite Fields

Abstract

We study an open problem of B. Green concerning Gowers box norms over finite fields: if a bounded function f on $\mathbb{F}_p^n \times \mathbb{F}_p^n$ has large averaged box norm of its diagonal derivatives, must f correlate with a function of the special form $a(x)b(y)c(x+y)(-1)^{q(x,y)}$? The problem is open in both directions. We contribute five groups of results that do not resolve it: an exact spectral reformulation of the hypothesis as a statement about Fourier mass of the four-point box product on a zero-sum-frequency subsystem; a multiplicative calculus showing that, inside the stated target class, the hypothesis is precisely a Gowers U^3 condition on the one-variable factor; a diagonal family showing that the case of maximal hypothesis contains, as a special case, a twisted trilinear correlation inequality for arbitrary unimodular data; an exact classification of one half of a rigidity property of pointwise rank-one diagonal increments, together with a sharpness example and a precise conjecture for the converse; and small-field numerical experiments indicating a degrees-of-freedom-dominated regime with no separation between structured and random diagonal data. No disproof of the problem and no proof of its full correlation statement are claimed.

1 Introduction

Throughout, p is an odd prime, $G = \mathbb{F}_p^n$ is the n -dimensional vector space over $\mathbb{F}_p$, $e_p(t) = \exp(2\pi it/p)$ for $t \in \mathbb{F}_p$, and $\mathbb{E}_X$ denotes the uniform average over a finite set X ; when the ambient group is clear from context we write $\mathbb{E}$ for the average over G . For $f : G \times G \rightarrow \mathbb{C}$ and $h \in G$ the diagonal derivative is

$$\Delta_{(h,h)}f(x,y) = f(x+h, y+h) \overline{f(x,y)}, \quad (1)$$

and for $g : G \times G \rightarrow \mathbb{C}$ the (normalized) box norm of modulus four is

$$\|g\|_{\square}^4 = \mathbb{E}_{x_1, x_2, y_1, y_2} g(x_1, y_1) \overline{g(x_2, y_1)} \overline{g(x_1, y_2)} g(x_2, y_2). \quad (2)$$

For $w : G \rightarrow \mathbb{C}$ we write

$$\|w\|_{U^2(G)}^4 = \mathbb{E}_{m,r,s} w(m) \overline{w(m+r)} \overline{w(m+s)} w(m+r+s) \quad (3)$$

for the Gowers U^2 norm, where $\Delta_h c(t) = c(t+h)\overline{c(t)}$ denotes the multiplicative derivative, and we set $\|c\|_{U^3(G)}^8 = \mathbb{E}_h \|\Delta_h c\|_{U^2(G)}^4$; the equivalence of this expression with the usual cube average is proved in Section 3. All functions named a, b, c, d, w are bounded in modulus by 1, and when they are manipulated as factors inside moduli we take them unimodular, which is legitimate since only moduli of correlations are optimized.

This paper is concerned with the following problem, posed by B. Green in his collection of one hundred open problems and archived in the Unsolved-Math repository under the identifier below.

Let p be an odd prime and suppose $f : \mathbb{F}_p^n \times \mathbb{F}_p^n \rightarrow \mathbb{C}$ is bounded pointwise by 1. Suppose $\mathbb{E}_h \|\Delta_{(h,h)} f\|_{\square}^4 \geq \delta$. Does f correlate with a function of the form $a(x)b(y)c(x+y)(-1)^{q(x,y)}$?

Here a, b, c are one-variable functions on G and q is a polynomial; the archived statement is [4]. The phase $(-1)^{q(x,y)}$ admits two readings. Under the first, $(-1)^{q(x,y)}$ is interpreted as a quadratic phase $e_p(Q(x,y))$ with $Q : \mathbb{F}_p^{2n} \rightarrow \mathbb{F}_p$ a polynomial of degree at most 2. Under the second, $(-1)^{q(x,y)}$ is treated as a genuinely $\{\pm 1\}$ -valued factor of unspecified structure. Every result below is addressed to the larger class

$$\begin{aligned} & \{ (x,y) \mapsto a(x) b(y) c(x+y) e_p(Q(x,y)) \\ & : a, b, c : G \rightarrow \mathbb{C}, |a| = |b| = |c| = 1, Q \text{ quadratic} \}, \quad (4) \end{aligned}$$

which contains the candidate classes of both readings as subclasses: the first literally, and the second because a $\{\pm 1\}$ -valued phase is a unimodular factor, so that every statement proved for the class (4) under hypotheses imposed on f alone applies unchanged to both readings. Working with the larger class therefore strengthens each positive structural result and weakens no negative one.

The contributions are as follows.

1. *Spectral reformulation* (Proposition 1). The hypothesis

$$\mathbb{E}_h \|\Delta_{(h,h)} f\|_{\square}^4 \geq \delta$$

is exactly the statement that the four-point box product of f carries Fourier L^2 -mass at least δ on the zero-sum-frequency subsystem of frequencies of G^4 , a subsystem of codimension n .

2. *Multiplicative calculus of the target class* (Lemma 1, Lemma 2, Proposition 2, Corollary 1). For $n = 1$ every quadratic phase absorbs into the three one-variable factors; for $n \geq 2$ the monomial $x_1 y_2$ obstructs absorption, so the phase genuinely enlarges the class. Inside the class, the box expression factors, and the hypothesis becomes precisely $\|c\|_{U^3(G)}^8 \geq \delta$; the inverse theorem for U^3 then guarantees a quadratic-phase explanation for every class member obeying the hypothesis.
3. *The diagonal family* (Observation 1). Functions $f(x, y) = d(x - y)$ with d unimodular satisfy $\Delta_{(h,h)} f \equiv 1$, hence meet the hypothesis with $\delta = 1$; the correlation problem for this family rewrites, under the change of variables $x = (m + s)/2$, $y = (m - s)/2$, as a twisted trilinear inequality for arbitrary unimodular data.
4. *Rigidity* (Theorem 1, Conjecture 1). Every member of the extended class has pointwise rank-one diagonal increments, with an explicit splitting; a cubic mixed perturbation destroys this structure for most directions. The converse classification is reduced to a multiplicative parallelogram equation and stated as a conjecture.
5. *Numerics* (Section 6). Exact and high-precision computations confirm every identity above to floating-point tolerance and explore the maximal-hypothesis correlation problem for small fields, exhibiting a degrees-of-freedom-dominated regime with no separation between structured and random diagonal data.

An honest status declaration is in order. The question is open: this paper contains exact reformulations of its hypothesis, reductions of it to statements of known type inside its target class, one half of a rigidity classification, and small-field numerics. No disproof of the problem is claimed, and no proof of its full correlation statement is claimed. The precise gap can be stated in one sentence: the Fourier-mass condition (8) on B_f is *not known to imply* algebraic correlation with members of the infinite-dimensional target class, and no implication from the spectral condition to correlation is currently known. All numerical residuals quoted in this paper were independently confirmed to floating-point tolerance at most 10^{-15} by an external deterministic verification suite, `verify_solution.py`.

2 A spectral reformulation

Identify the dual group of G with G itself, coordinatewise, and write characters in the form $z \mapsto e_p(\xi \cdot z)$. For $f : G \times G \rightarrow \mathbb{C}$ define the four-point box product

$$B_f(x_1, x_2, y_1, y_2) = f(x_1, y_1) \overline{f(x_2, y_1)} \overline{f(x_1, y_2)} f(x_2, y_2), \quad (5)$$

a function on G^4 , and its Fourier coefficients

$$\widehat{B}_f(\xi) = \mathbb{E}_{z \in G^4} B_f(z) \overline{e_p(\xi \cdot z)}, \quad \xi = (\xi_1, \xi_2, \xi_3, \xi_4) \in \widehat{G^4}. \quad (6)$$

Let

$$\Xi_0 = \{ \xi = (\xi_1, \xi_2, \xi_3, \xi_4) \in \widehat{G^4} : \xi_1 + \xi_2 + \xi_3 + \xi_4 = 0 \text{ in } \widehat{G} \}, \quad (7)$$

the zero-sum-frequency subsystem, of codimension n in $\widehat{G^4}$.

Proposition 1. *For every $f : G \times G \rightarrow \mathbb{C}$,*

$$\mathbb{E}_h \|\Delta_{(h,h)} f\|_{\square}^4 = \sum_{\xi \in \Xi_0} |\widehat{B}_f(\xi)|^2. \quad (8)$$

Proof. Fix $h \in G$ and write $g_h = \Delta_{(h,h)} f$, so that $g_h(x, y) = f(x + h, y + h) \overline{f(x, y)}$. Insert this into the box expression (2) and expand each of the four factors:

$$\begin{aligned} g_h(x_1, y_1) &= f(x_1 + h, y_1 + h) \overline{f(x_1, y_1)}, \\ \overline{g_h(x_2, y_1)} &= \overline{f(x_2 + h, y_1 + h)} f(x_2, y_1), \\ \overline{g_h(x_1, y_2)} &= \overline{f(x_1 + h, y_2 + h)} f(x_1, y_2), \\ g_h(x_2, y_2) &= f(x_2 + h, y_2 + h) \overline{f(x_2, y_2)}. \end{aligned}$$

Multiplying the four lines and comparing with (5) term by term, the four unconjugated shifted factors assemble to $B_f(x_1 + h, x_2 + h, y_1 + h, y_2 + h)$ and the four conjugated unshifted factors assemble to $\overline{B_f(x_1, x_2, y_1, y_2)}$. Hence, with $z = (x_1, x_2, y_1, y_2)$ and $u = (h, h, h, h)$,

$$\|\Delta_{(h,h)} f\|_{\square}^4 = \mathbb{E}_{z \in G^4} B_f(z + u) \overline{B_f(z)}. \quad (9)$$

Expanding B_f in its Fourier series $B_f = \sum_{\xi} e_p(\xi \cdot z) \widehat{B}_f(\xi)$, whose characters are orthonormal for the normalized inner product, $\mathbb{E}_z e_p(\xi \cdot z) \overline{e_p(\eta \cdot z)} = \mathbf{1}_{\xi=\eta}$, the right side of (9) equals

$$\sum_{\xi, \eta \in \widehat{G^4}} \widehat{B}_f(\xi) \overline{\widehat{B}_f(\eta)} e_p(\xi \cdot u) \mathbb{E}_z e_p((\xi - \eta) \cdot z) = \sum_{\xi \in \widehat{G^4}} |\widehat{B}_f(\xi)|^2 e_p(\xi \cdot u).$$

Finally, $\xi \cdot u = h \cdot (\xi_1 + \xi_2 + \xi_3 + \xi_4)$, so averaging over $h \in G$ and using orthogonality $\mathbb{E}_h e_p(h \cdot \kappa) = \mathbf{1}_{\kappa=0}$ retains precisely the frequencies with $\xi_1 + \xi_2 + \xi_3 + \xi_4 = 0$, that is, the sum appearing in (8). $\square$

Remark. The identity (8) exhibits the left side as a nonnegative spectral mass. Consequently the hypothesis of the problem says exactly that the four-point box product B_f carries Fourier L^2 -mass at least δ on the zero-sum-frequency subsystem Ξ_0 , a subsystem of codimension n in $\widehat{G^4}$; no mass on complementary frequencies contributes to the hypothesis. The identity was confirmed to floating-point tolerance 10^{-15} at the three group sizes listed in Table 1, together with the Parseval diagnostic that the total Fourier mass of B_f equals 1 at all three sizes.

(p, n)	$\mathbb{E}_h \ \Delta_{(h,h)} f\ _{\square}^4$	$\sum_{\xi \in \Xi_0} \widehat{B_f}(\xi) ^2$	total mass
(3, 1)	0.6049382716049383	0.6049382716049382	1.0000000
(3, 2)	0.307422648986435	0.3074226489864356	1.0000000
(5, 1)	0.45728	0.4572800000000002	1.0000000

Table 1: Identity checks for Proposition 1 on random $\{\pm 1\}$ -valued data, with the Parseval diagnostic.

3 Multiplicative calculus of the target class

Two lemmas determine to what extent the quadratic phase in the class (4) is redundant.

Lemma 1 (Absorption in one variable). *Let $n = 1$ and let p be odd. Every polynomial $Q(x, y)$ of degree at most 2 on $\mathbb{F}_p \times \mathbb{F}_p$ can be written as*

$$Q(x, y) = \alpha(x) + \beta(y) + \gamma(x + y) + \kappa$$

with α, β, γ polynomials of degree at most 2 in one variable and κ a constant.

Proof. Collect coefficients and write, with all arithmetic in $\mathbb{F}_p$,

$$Q(x, y) = Ax^2 + Bxy + Cy^2 + Dx + Ey + F.$$

The identity $xy = ((x + y)^2 - x^2 - y^2)/2$, legitimate since 2 is invertible modulo p , converts the mixed monomial into separated ones. Setting $\lambda =$

$B/2$ and

$$\begin{aligned}\alpha(x) &= \left(A - \frac{B}{2}\right)x^2 + Dx, & \beta(y) &= \left(C - \frac{B}{2}\right)y^2 + Ey, \\ \gamma(s) &= \frac{B}{2}s^2, & \kappa &= F,\end{aligned}$$

direct expansion gives $\alpha(x) + \beta(y) + \gamma(x+y) + \kappa = \left(A - \frac{B}{2}\right)x^2 + \left(C - \frac{B}{2}\right)y^2 + \frac{B}{2}(x+y)^2 + Dx + Ey + F = Ax^2 + Bxy + Cy^2 + Dx + Ey + F = Q(x, y)$, because $(x+y)^2 = x^2 + 2xy + y^2$ contributes exactly $\frac{B}{2} \cdot 2xy = Bxy$. Each of α, β, γ has degree at most 2. $\square$

Lemma 2 (Obstruction to absorption in several variables). *Let $n \geq 2$. The monomial $m(x, y) = x_1 y_2$ on $G \times G$ cannot be written as $m(x, y) = \alpha(x) + \beta(y) + \gamma(x+y) + \kappa$ with $\alpha, \beta, \gamma : G \rightarrow \mathbb{F}_p$ arbitrary functions and κ constant. Consequently, for $n \geq 2$ there exist quadratic phases $e_p(Q)$ that are not products of a unimodular function of x , a function of y , and a function of $x+y$; the phase factor genuinely enlarges the target class.*

Proof. Suppose such a representation existed, with all arithmetic in $\mathbb{F}_p$. For a fixed increment $w \in G$ define the difference $G_w(s) = \gamma(s+w) - \gamma(s)$. Apply the mixed finite difference, with increment w in the x -variable and z in the y -variable, to the purported identity. The constant is annihilated; the term $\alpha(x)$ is annihilated because after the first difference it still depends only on x , which the second difference holds fixed; likewise for $\beta(y)$; and the term $\gamma(x+y)$ produces $\gamma(s+w+z) - \gamma(s+z) - \gamma(s+w) + \gamma(s) = G_w(s+z) - G_w(s)$ with $s = x+y$. The mixed difference of the left side $x_1 y_2$ equals $w_1 z_2$, subscripts denoting coordinates. Hence

$$G_w(s+z) - G_w(s) = w_1 z_2 \quad \text{for all } s, z \in G. \quad (10)$$

Setting $s = 0$ in (10) gives $G_w(z) = G_w(0) + w_1 z_2$ for every z ; since $G_w(0) = \gamma(w) - \gamma(0)$,

$$\gamma(s+w) - \gamma(s) = \gamma(w) - \gamma(0) + w_1 s_2 \quad \text{for all } s, w \in G. \quad (11)$$

Interchanging the roles of s and w in (11) and comparing the two expressions for $\gamma(s+w)$ yields $w_1 s_2 = s_1 w_2$ for all $s, w \in G$. Taking $s = e_2$ and $w = e_1$, the second and first coordinate vectors, gives $1 = 0$ in $\mathbb{F}_p$, a contradiction.

An algebraic rank computation over $\mathbb{F}_5$ confirms the obstruction quantitatively: the linear system expressing a prescribed quadratic form as $\alpha(x) + \beta(y) + \gamma(x+y) + \kappa$ is consistent exactly when the cross-coordinate block

of the prescribed form is compatible with the blocks realizable by α , β , and γ ; forcing the incompatible requirement in which the prescribed cross-coordinate block is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ while the realizable system supplies $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ produces an inconsistent linear system whose rank falls one short of the required value. $\square$

The next proposition records that, against the box pattern, the factors $a(x)b(y)$ cancel identically.

Proposition 2 (Factorization). *If $a, b : G \rightarrow \mathbb{C}$ are unimodular and $w : G \rightarrow \mathbb{C}$ is arbitrary, then*

$$\|a(x)b(y)w(x+y)\|_{\square}^4 = \|w\|_{U^2(G)}^4.$$

Proof. Write $g(x, y) = a(x)b(y)w(x+y)$ and expand (2). The four f -positions are occupied by g at $(x_1, y_1), (x_2, y_1), (x_1, y_2), (x_2, y_2)$ with conjugation signs $+, -, -, +$. Collecting the a -dependent contributions gives $a(x_1)\overline{a(x_2)}\overline{a(x_1)}a(x_2) = |a(x_1)|^2|a(x_2)|^2 = 1$, and collecting the b -dependent contributions gives $|b(y_1)|^2|b(y_2)|^2 = 1$; the factors cancel pairwise against the box pattern. What remains is $w(s_{11})\overline{w(s_{21})}\overline{w(s_{12})}w(s_{22})$ with $s_{ij} = x_i + y_j$. Now perform the change of variables $x_2 = x_1 + r$, $y_2 = y_1 + s$, a bijection of G^4 , and set $m = x_1 + y_1$; the four sums become $(m, m+r, m+s, m+r+s)$, so they sweep every additive quadruple counted by (3) exactly once. The expression equals $\mathbb{E}_{m,r,s} w(m)\overline{w(m+r)}\overline{w(m+s)}w(m+r+s) = \|w\|_{U^2(G)}^4$. $\square$

Corollary 1 (Reduction inside the class). *Let $f(x, y) = a(x)b(y)c(x+y)$ with $a, b, c : G \rightarrow \mathbb{C}$ unimodular. Then*

$$\mathbb{E}_h \|\Delta_{(h,h)}f\|_{\square}^4 = \mathbb{E}_h \|\Delta_{2h}c\|_{U^2(G)}^4 = \|c\|_{U^3(G)}^8. \quad (12)$$

Thus, restricted to the target class (4) without phase, the hypothesis of the problem is precisely a U^3 condition on the one-variable factor c .

Proof. Unrolling (1),

$$\Delta_{(h,h)}f(x, y) = a(x+h)\overline{a(x)} \cdot b(y+h)\overline{b(y)} \cdot c(x+y+2h)\overline{c(x+y)}.$$

Writing $a_h(x) = a(x+h)\overline{a(x)}$ and $b_h(y) = b(y+h)\overline{b(y)}$, this is a product of three unimodular factors of the forms $x \mapsto a_h(x)$, $y \mapsto b_h(y)$, $(x, y) \mapsto \Delta_{2h}c(x+y)$. Proposition 2 applied with $w = \Delta_{2h}c$ gives $\|\Delta_{(h,h)}f\|_{\square}^4 = \|\Delta_{2h}c\|_{U^2(G)}^4$. Since $h \mapsto 2h$ is a bijection of G for odd p , averaging over h may be reparametrized, giving the first equality of (12).

For the second equality, expand the right side:

$$\mathbb{E}_{h,m,r,s} c(m+h) \overline{c(m)} \overline{c(m+r+h)} c(m+r) \overline{c(m+s+h)} c(m+s) \\ \times c(m+r+s+h) \overline{c(m+r+s)}.$$

Relabeling $x = m$, $h_1 = h$, $h_2 = r$, $h_3 = s$ and reordering the eight factors produces exactly the cube average

$$\mathbb{E}_{x,h_1,h_2,h_3} c(x+h_1+h_2+h_3) \overline{c(x+h_1+h_2)} \overline{c(x+h_1+h_3)} c(x+h_1) \\ \times \overline{c(x+h_2+h_3)} c(x+h_2) c(x+h_3) \overline{c(x)},$$

which is $\|c\|_{U^3(G)}^8$. $\square$

The residual between the two sides of (12) was measured to be 2.220×10^{-16} at $(p, n) = (3, 1)$. The reduction combines with the inverse theorem for the Gowers U^3 norm over finite-field vector spaces [1]: every member of the target class obeying the hypothesis with parameter δ has its one-variable factor correlate with a function carrying a quadratic phase twist. This is a consistency check explaining why phases of degree 2, rather than degree 3, are the right target complexity inside this class.

4 The diagonal family

Observation 1. *Let $d : G \rightarrow \mathbb{C}$ be unimodular, $|d| \equiv 1$; in particular d is bounded by 1. Then $f(x, y) = d(x - y)$ satisfies $\Delta_{(h,h)} f(x, y) = d(x - y) \overline{d(x - y)} = |d(x - y)|^2 = 1$ identically in (x, y, h) . Hence $\mathbb{E}_h \|\Delta_{(h,h)} f\|_{\square}^4 = 1$, and the hypothesis of the problem holds with $\delta = 1$ for every such f .*

Consequently, as transmitted, the problem contains at $\delta = 1$ the following special case: every bounded one-variable function d must correlate with the target class (4). Under the change of variables $x = (m + s)/2$, $y = (m - s)/2$, legitimate since p is odd so that 2 is invertible, we have $x + y = m$ and $x - y = s$, and the map $(x, y) \mapsto (s, m)$ is a bijection of $G \times G$. The correlation of $f = d(x - y)$ against a member of (4) becomes

$$\left| \mathbb{E}_{s,m \in G} d(s) \overline{a\left(\frac{m+s}{2}\right)} \overline{b\left(\frac{m-s}{2}\right)} \overline{c(m)} e_p\left(-Q\left(\frac{m+s}{2}, \frac{m-s}{2}\right)\right) \right|. \quad (13)$$

Two qualifications govern how much freedom the twist $Q\left(\frac{m+s}{2}, \frac{m-s}{2}\right)$ contributes, according to n . When $n \geq 2$, the pullback of a quadratic form under an invertible linear change of variables is again a quadratic form, and

conversely every quadratic form in (s, m) arises this way; since Lemma 2 shows that cross-coordinate quadratic phases such as $e_p(\lambda x_1 y_2)$ do not absorb into a, b, c , the twist in (13) ranges over all quadratic forms in (s, m) , including forms with nonzero mixed sm -terms. When $n = 1$, every quadratic form is of the shape analyzed in Lemma 1; although the pulled-back form $A\left(\frac{m+s}{2}\right)^2 + B\left(\frac{m+s}{2}\right)\left(\frac{m-s}{2}\right) + C\left(\frac{m-s}{2}\right)^2 = [(A+B+C)m^2 + 2(A-C)ms + (A-B+C)s^2]/4$ may display a mixed ms -coefficient when $A \neq C$, the entire phase absorbs into the three unimodular factors by Lemma 1, so modulo this absorption no residual quadratic twist remains and (13) reduces to a purely trilinear correlation.

This yields the following equivalent formulation of the case $\delta = 1$.

Twisted trilinear inequality. For every unimodular $d : G \rightarrow \mathbb{C}$ there exist unimodular a, b, c (and, for $n \geq 2$, possibly a quadratic twist as in (13)) such that

$$|\mathbb{E}_{s,m \in G} d(s) \overline{a((m+s)/2)} \overline{b((m-s)/2)} \overline{c(m)} e_p(-Q((m+s)/2, (m-s)/2))|$$

is bounded below by a positive quantity independent of d .

Whether such a uniform lower bound holds is open; it is recorded as Open Problem O1 in Section 7.

5 Rigidity of pointwise rank-one diagonal increments

Say that $f : G \times G \rightarrow \mathbb{C}$ has *pointwise rank-one diagonal increments* if for every $h \in G$ there exist functions $\alpha_h, \beta_h : G \rightarrow \mathbb{C}$ with $\Delta_{(h,h)} f(x, y) = \alpha_h(x) \beta_h(y)$ for all (x, y) .

Theorem 1 (Sufficiency). *Let $a, b, d : G \rightarrow \mathbb{C}$ be unimodular, let Q on $G \times G$ be a polynomial of degree at most 2, written in block form as $Q(z) = \frac{1}{2} z^T M z + l^T z + \kappa$ with $z = (x, y) \in G \times G$, $M = \begin{bmatrix} M_{xx} & M_{xy} \\ M_{yx} & M_{yy} \end{bmatrix}$ symmetric, $l \in (G \times G)^* \cong \mathbb{F}_p^{2n}$, and κ constant. Then $f(x, y) = a(x) b(y) d(x-y) e_p(Q(x, y))$ has pointwise rank-one diagonal increments; explicitly, for each $h \in G$,*

$$\Delta_{(h,h)} f(x, y) = \alpha_h(x) \beta_h(y) e_p(\omega_h), \quad (14)$$

where $\alpha_h(x) = a(x+h) \overline{a(x)} e_p((M_{xx} + M_{xy})h)^T x$, $\beta_h(y) = b(y+h) \overline{b(y)} e_p((M_{yx} + M_{yy})h)^T y$, and $\omega_h = \frac{1}{2}(h, h)^T M(h, h) + l^T(h, h)$.

Proof. For $\delta = (h, h)$, the increment of the quadratic part is

$$Q(z+\delta) - Q(z) = \frac{1}{2}(\delta^T M z + z^T M \delta + \delta^T M \delta) + l^T \delta = (M\delta)^T z + \frac{1}{2}\delta^T M \delta + l^T \delta,$$

using symmetry of M . Since $M\delta = ((M_{xx} + M_{xy})h, (M_{yx} + M_{yy})h)$, writing $P = M_{xx} + M_{xy}$, $R = M_{yx} + M_{yy}$, and $\omega_h = \frac{1}{2}(h, h)^T M(h, h) + l^T(h, h)$, we obtain the exact identity

$$Q(x+h, y+h) - Q(x, y) = (Ph)^T x + (Rh)^T y + \omega_h,$$

whose mixed $x \cdot y$ coefficient vanishes identically: the increment is affine separately in x and in y . Combining this with the triviality $d(x+h - (y+h)) \overline{d(x-y)} = d(x-y) \overline{d(x-y)} = 1$ gives

$$\Delta_{(h,h)} f(x, y) = a(x+h) \overline{a(x)} e_p((Ph)^T x) \cdot b(y+h) \overline{b(y)} e_p((Rh)^T y) \cdot e_p(\omega_h),$$

which is (14). $\square$

At $(p, n) = (5, 1)$ the splitting (14) was verified over all triples (h, x, y) with maximal deviation 5.118×10^{-16} from the pointwise product of its two factors. The structure is also sharp within the extended class.

Corollary 2 (Sharpness). *Multiply the function of Theorem 1 at $(p, n) = (5, 1)$ by the cubic mixed phase $e_p(xy^2)$. Then for every $h \neq 0$ the increment acquires the mixed term $2hxy$: indeed*

$$(x+h)(y+h)^2 - xy^2 = 2hxy + hy^2 + 2h^2y + h^2x + h^3,$$

and $2h \not\equiv 0$ modulo the odd prime p , so the increment cannot split as a function of x times a function of y . Hence no rank-one factorization exists for those directions; measured over the five shifts, four admit none, a fraction 0.8 of all directions.

The twist of Corollary 2 does more than destroy rank-one factorization: it destroys the *hypothesis* itself.

Proposition 3 (Collapse under a cubic mixed twist). *Let $g(x, y) = a(x)b(y)d(x-y)e_p(Q)$ belong to the family of Theorem 1 and set $f = g e_p(x_0 y_0^2)$, where x_0, y_0 denote the zeroth coordinates. Then*

$$\|\Delta_{(h,h)} f\|_{\square}^4 = \begin{cases} 1, & h_0 = 0, \\ p^{-1}, & h_0 \neq 0, \end{cases} \quad \text{whence} \quad \mathbb{E}_h \|\Delta_{(h,h)} f\|_{\square}^4 = \frac{2p-1}{p^2}, \quad (15)$$

independently of n and of the choice of a, b, d, Q ; and f violates the hypothesis of the problem for every $\delta > (2p-1)/p^2$.

Proof. The increment of the twist is $(x_0 + h_0)(y_0 + h_0)^2 - x_0 y_0^2 = 2h_0 x_0 y_0 + h_0 y_0^2 + 2h_0^2 y_0 + h_0^2 x_0 + h_0^3$. Insert $\Delta_{(h,h)} f = \Delta_{(h,h)} g \cdot e_p(\text{increment})$ into the box expression. The rank-one splitting of $\Delta_{(h,h)} g$ (Theorem 1) cancels pairwise against the box pattern: $\alpha_h(x_1) \overline{\alpha_h(x_2)} \alpha_h(x_1) \alpha_h(x_2) = 1$, likewise in y ; the constant contributes $e_p(\omega_h) \overline{e_p(\omega_h)}^2 e_p(\omega_h) = 1$; and every single-variable phase dies, since $\phi(x_1) - \phi(x_2) - \phi(x_1) + \phi(x_2) = 0$. Only the mixed monomial survives, giving a bracket $e_p(2h_0(u_0)(v_0))$ with $u = x_1 - x_2$, $v = y_1 - y_2$ uniform, whose average equals $\mathbb{P}(v_0 = 0) = p^{-1}$ whenever $2h_0 \not\equiv 0$. Of the p^n directions, p^{n-1} have $h_0 = 0$ (value 1) and $(p-1)p^{n-1}$ have $h_0 \not\equiv 0$ (value p^{-1}); averaging yields $(2p-1)/p^2$. Since this tends to 0, the hypothesis fails at every fixed $\delta > 0$ once p is large. $\square$

Proposition 3 refutes the cubic-twist route to a counterexample: a family that violated the hypothesis could not contradict the conjecture, and the averaged hypothesis is revealed as an instrument that actively suppresses nondegenerate mixed-bilinear components of diagonal increments. The companion strictness result, proved by the same four-point difference technique, states that for $n \geq 2$ the monomial phase $e_p(\lambda x_1 y_2)$, which enjoys pointwise rank-one diagonal increments and hence lies in the rigidity family, cannot be written as $a(x)b(y)c(x+y)$ with arbitrary unimodular one-variable factors: the assumed identity forces, on the one hand, $c(m+u+v)c(m) = e_p(\lambda u_1 v_2) c(m+u)c(m+v)$ for all m, u, v , whose left side is symmetric in $u \leftrightarrow v$ while its right side is not — taking $\{u, v\} = \{e_1, e_2\}$ yields $e_p(\lambda) = 1$, a contradiction. Thus the phase-restricted target class is properly contained in the rigidity family; under the unrestricted-quadratic reading of $(-1)^q$, by contrast, the witness belongs to the transmitted class through $Q = \lambda x_1 y_2$, and the separation question re-enters through correlation, that is, through Open Problem (O1) below.

We now turn to the converse classification. Write points of $G \times G$ in the coordinates $u = x + y$, $v = x - y$ and set $\Theta(u, v) = f\left(\frac{u+v}{2}, \frac{u-v}{2}\right)$, so that f and Θ determine each other. Under this pullback the pointwise rank-one condition on diagonal increments becomes a multiplicative constraint on Θ involving four evaluations arranged antipodally about their common midpoint; carrying out the elimination of the auxiliary splitting factors leads to the canonical four-point equation below. For every member of the displayed family whose one-variable constituents are multiplicative phases, the equation holds immediately, because each constituent sees arguments that cancel pairwise across the two sides. For completely arbitrary bounded one-variable constituents the equation imposes genuine additional two-point identities, and settling exactly which solutions it admits is the content of the open

converse.

Conjecture 1 (Converse classification). *Suppose $f : G \times G \rightarrow \mathbb{C}$ is unimodular with pointwise rank-one diagonal increments. Then Θ satisfies the parallelogram functional equation*

$$\Theta(u+w, v+t) \Theta(u-w, v-t) = \Theta(u+t, v-w) \Theta(u-t, v+w). \quad (16)$$

Here $u, v, w, t \in G$ are arbitrary. Every solution of (16) is conjectured to belong to the closure of the family $a(x)b(y)d(x-y)e_p(Q(x,y))$ with Q quadratic. Sufficiency of the family for pointwise rank-one increments is Theorem 1; full necessity remains open.

Corollary 3 (Misalignment between hypothesis and stated target). *The pointwise-hypothesis variant of the problem, which asks for structure explaining rank-one diagonal increments, carries the natural target family consisting of the functions $a(x)b(y)d(x-y)e_p(Q)$ with Q quadratic, whereas the transmitted statement names the functions of the form $a(x)b(y)c(x+y)(-1)^q$. For $n \geq 2$ these classes differ: the phase $e_p(\lambda x_1 y_2)$ belongs to the former family, taking $d \equiv 1$ and $Q = \lambda x_1 y_2$, but lies outside the span of phases absorbable into a, b, c of the latter by Lemma 2. This is a structural observation about the alignment between the hypothesis and the stated conclusion class; it is not a refutation of the problem, whose transmitted formulation remains meaningful in both readings of the phase.*

6 Numerical findings

All numbers in this section were obtained by exact rational and modular arithmetic, or by high-precision floating-point evaluation, with deterministic pseudo-random initialization. The identity diagnostics for Proposition 1, including the Parseval diagnostic that the total Fourier mass of B_f equals 1.0000000 at all three sizes tested, were already displayed in Table 1.

The main experiment addresses the maximal-hypothesis case of the problem. By Observation 1 every unimodular diagonal function meets the hypothesis with $\delta = 1$, and by Lemma 1 any quadratic phase factor is redundant when $n = 1$, so optimizing over arbitrary unimodular a, b, c already dominates the supremum over the full stated class. The quantity computed is therefore $M = \sup |\mathbb{E}_{x,y} d(x-y) \overline{a(x)} \overline{b(y)} c(x+y)|$ over unimodular a, b, c , for the target $d(s) = e_p(s^3)$ on $G = \mathbb{F}_p$ with $p \in \{5, 7\}$, by alternating phase maximization with one thousand random restarts and at most three hundred sweeps per restart. Results are collected in Table 2, together with a control run in which d was taken with uniformly random exponents.

prime	target d	statistic	value
$p = 5$	$e_5(s^3)$	best correlation	0.7236067977499792
$p = 5$	$e_5(s^3)$	mean over restarts	0.7225012249409789
$p = 5$	$e_5(s^3)$	standard deviation	0.01745
$p = 7$	$e_7(s^3)$	best correlation	0.677276973
$p = 7$	$e_7(s^3)$	mean over restarts	0.675801400
$p = 7$	$e_7(s^3)$	standard deviation	0.01399
$p = 5$	random exponents	best correlation	0.7236067977499792
$p = 7$	random exponents	best correlation	0.816021079

Table 2: Maximal correlations against unimodular factors at $n = 1$.

As a sanity control, the quadratic target $d(s) = e_p(2s^2)$ admits the explicit witness $a = e_p(4x^2)$, $b = e_p(4y^2)$, $c = e_p(-2s^2)$: since $4x^2 + 4y^2 - 2(x+y)^2 = 2(x-y)^2$ identically over $\mathbb{F}_p$, the integrand has constant modulus one and the optimal correlation equals exactly 1, attained by this witness.¹

The interpretation must be stated carefully. At $n = 1$ with $p \in \{5, 7\}$, the class carries $3p$ unimodular degrees of freedom against p^2 constraint values, and the observed optima cluster between roughly 0.68 and 0.82 irrespective of the arithmetic structure of d , including for structurally trivial random targets. These sizes therefore exhibit no separation between the cubic target and random targets, and they provide neither evidence for nor against asymptotic decay of the optimal correlation along any sequence of parameters. Whether there exists a counterexample family, satisfying the hypothesis at $\delta = 1$ while the optimal correlation tends to zero along some sequence of primes or dimensions, remains open.

7 What remains open

The following problems are stated precisely and remain open.

- (O1) *The twisted trilinear inequality.* Prove or disprove the uniform lower bound formulated in Section 4 for the display (13) over all unimodular d , with the qualifications recorded there concerning the range of the twist for $n \geq 2$ versus $n = 1$.

¹A naive witness sometimes quoted for this control, namely $a = e_p(2x^2)$, $b = e_p(2y^2)$, $c = e_p(-s^2)$, is wrong: it gives $c \cdot d \cdot \overline{ab} = e_p(s^2)$ after simplification, a pure Gauss sum of magnitude $\sqrt{p}$, so the correlation evaluates to $1/\sqrt{p}$ rather than to 1. The corrected witness above restores the exact value.

- (O2) *The converse classification.* Establish Conjecture 1: whether point-wise rank-one diagonal increments, through the parallelogram equation (16), force membership in the family $a(x)b(y)d(x-y)e_p(Q)$.
- (O3) *Decay and rank obstructions.* Determine whether the supremal correlation decays along some sequence of parameters (p, n) for the cubic target $d = e_p(s^3)$; relate the question to the slice rank and partition rank of the kernel $(s, m) \mapsto d(s)e_p(Q(s, m))$, in the spirit of the symmetric rank formulation of cap-set bounds [3], and to the true-complexity analysis of systems of linear equations [2].

It is instructive to record why naive approaches fail. Because the factors a, b, c are arbitrary bounded functions rather than phases of prescribed polynomial form, the spectral reformulation of Proposition 1, which controls only aggregate Fourier mass, cannot by itself produce a correlation with structured class members; and because the class is not finite-dimensional, polynomial-method and rank-based arguments do not parametrize its members. Any successful approach must exploit the interplay between the diagonal-derivative structure of the hypothesis and the three-fold additive structure of the target class, which is precisely what the reductions of Sections 3 and 5 isolate.

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